

15. Given that $\sin(A) = \frac{2}{5}$ and $\frac{\pi}{2} < A < \pi$ (where A is measured in radians). Then $\sin(2A)$ can be expressed in simplest radical form as $\frac{k\sqrt{w}}{p}$ where k and p are relatively prime integers and $p > 0$. Find the value of $(k + w + p)$.
16. From a group of 8 men and 7 women, a committee of 3 is selected at random. Find the probability that the committee consists of 2 members of one sex and 1 member of the opposite sex.
17. If $\sin(x) = \frac{5}{13}$ and $\cot(y) = -\frac{3}{4}$, then there are two possible answers for $\tan(x - y)$. Find the sum of these two possible distinct answers. Express your answer as an improper fraction reduced to lowest terms.
18. Let $C(n, k) = \frac{n!}{k!(n-k)!}$ where n and k represent positive integers. Find the **sum** of all distinct values of n such that $97 < C(n, 6) < 2114$.
19. The graph of the rational function $g(x) = \frac{x^2 - 5x + 6}{x - 4}$ has a slant (oblique) asymptote which can be expressed in the form: $f(x) = mx + b$. Find the value of $(m + b)$.
20. Let x and n represent two digit positive **odd** integers for which the tens digit is non-zero and for which $n > x$. Find the largest possible value for x for which the sum of the consecutive positive **odd** integers from 1 through x is 99 less than the sum of the consecutive positive **odd** integers from x through n .